

Documentation interfaces

This document contains the description of the interfaces used in the file “main_interfaz.f90”.

Note: the variables that contain an asterisk are **optional** arguments.

read_chemistry_interface

- **Description:** Subroutine that reads all the chemical input data (species, reactions, initial and external concentrations, mineral zones, etc.) by dispatching to the appropriate format-specific reader.
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
root	String	Root of input and output files of problem to be solved
path_pb	String	Path to input and output files of problem to be solved
path_DB	String	Path to chemical databases
num_tar	Integer	Expected number of target (domain) waters from the mesh

- **Pre-process:**
 - Sets arbitrary file units for the chemical system, computation options and local chemistry input files.
- **Process:**
 - Selects the input format option (read_opt) and reads the **chemical data** accordingly. Currently only the CHEPROO-based format (read_opt = 1) is fully implemented; the PHREEQC (read_opt = 2) and PFLOTRAN (read_opt = 3) branches abort execution with an error stop.

write_conc_comp_wat_types

- **Description:** Subroutine that writes the concentrations of variable activity species and aqueous components of every water type to a file. A reference water type whose name contains **domain**, **initial**, **resident** (or their Spanish equivalents) is located, and its primary-species basis is used to project the variable-activity-species concentrations of every water type onto a common component vector, so all water types yield component vectors of identical dimension. Water types are split into an initial group (matching the reference name pattern) and an external group (all others), and each group is written separately.
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
path	String	Path to the output file
filename	String	Output file name

- **Process:**
 - Locates the reference water type by case-insensitive name matching (**domain / initial / resident / dominio / inicial / residente**) and aborts with an error stop if none is found.
 - Classifies water types into two groups by name: initial (those whose name matches the same domain / initial / resident set used for the reference) and external (everything else).
 - Opens the output file (path//filename) and writes a header with the number of aqueous components.
 - Writes the list of variable-activity aqueous species of the reference water type, with primary species in rows 1 to num_aq_prim_species followed by secondary variable-activity species.
 - Writes the aqueous components as linear combinations of the variable-activity species defined by comp_mat_aq, for example $u_6 = \text{hco}_3^- + \text{co}_3^{2-}$.
 - For each initial water type, projects its variable-activity-species concentrations onto the reference basis and writes two scientific-notation tables: variable-activity species concentrations (primary rows first, then secondary) and aqueous component concentrations.
 - For each external water type, performs the same projection and writes the corresponding variable-activity species and aqueous component concentration tables.
 - Closes filename

get_num_aq_comps_tar_wat

- **Description:** Function that returns the number of aqueous components in a target water. If no index is provided, the first target water is used.
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
ind_tw*	Integer	Optional index of a target water

- **Output:**

Variable	Type	Description
num_aq_comps	Integer	Number of aqueous components in the target water

- **Process:**
 - Returns the value of num_aq_prim_species in the speciation_alg object associated to the target water.

interfaz_comps_arch_eq

- **Description:** Subroutine that solves a reactive mixing iteration with **components**, and writes aqueous component and variable activity species concentrations, using **files** as arguments. The integration method is **Euler explicit** and the reaction mixing ratios are **lumped**. It assumes that there are equilibrium reactions only (no kinetics), so the aqueous component concentrations after reactive mixing are identical to those after conservative transport.
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
path	String	Path to input and output files.
num_aq_comps	Integer	Nº aqueous components.
file_in	String	File containing concentrations of aqueous components after solving a conservative transport iteration ($\tilde{\mathbf{U}}^{k+1}$).
Delta_t	Real	Time step (Δt^k)
file_out	String	File where aqueous component concentrations ($\mathbf{U}^{k+1}$) and variable activity species concentrations ($\mathbf{C}_v^{k+1}$) after solving a reactive mixing iteration will be written.

- **Pre-process:**
 - Allocate aqueous component concentrations after conservative transport ($\tilde{\mathbf{U}}^{k+1}$)
 - Determine, from the first target water, the number of variable activity species `n_nc`.
 - Allocate the variable activity species concentrations `conc_nc` with shape (n_nc, num_target_waters).
- **Process:**
 - **Read $\tilde{\mathbf{U}}^{k+1}$ from file_in.** **Note:** The rows must represent components and the columns targets.
 - **Solve** reactive mixing iteration for each target water j :
 - **Loop over number of target waters**
 - Speciate to compute variable activity species concentrations $\mathbf{c}_{v,j}^{k+1}$ from `u_tilde`, using Newton-Raphson.
 - Check Newton convergence: if speciation does not converge for a target water, the target index and a "No convergence in speciation after reactive mixing iteration. Try reducing the time step" message are printed and execution is aborted with an error stop.

- **Post-process:**

- **Write $\mathbf{U}^{k+1}$ in file_out** (same dimensions as $\tilde{\mathbf{U}}^{k+1}$)
- **Write $\mathbf{c}_v^{k+1}$ in file_out** (rows: species, columns: targets)
- **Deallocate arrays**

interfaz_comps_arch_eq_kin

- **Description:** Subroutine that solves a reactive mixing iteration with **components**, and writes aqueous component concentrations, variable activity species concentrations and kinetic reaction rates, using **files** as arguments. The integration method is **Euler explicit** and the reaction mixing ratios are **lumped**. It assumes that there are both equilibrium and kinetic reactions.
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
path	String	Path to input and output files.
num_aq_comps	Integer	Nº aqueous components.
file_in	String	File containing concentrations of aqueous components after solving a conservative transport iteration ($\tilde{\mathbf{U}}^{k+1}$).
Delta_t	Real	Time step (Δt^k)
file_out	String	File where aqueous component concentrations ($\mathbf{U}^{k+1}$) and variable activity species concentrations ($\mathbf{C}_v^{k+1}$) after solving a reactive mixing iteration will be written.

- **Pre-process:**
 - Allocate aqueous component concentrations after conservative transport ($\tilde{\mathbf{U}}^{k+1}$).
 - Allocate chemical reactions contribution to aqueous component concentrations ($\mathbf{U}_K^{k+1}$).
 - Determine, from the first target water, the number of kinetic reactions (aqueous kinetic n_aq_kin, mineral kinetic n_min_kin) and the number of variable activity species n_nc.
 - Allocate the variable activity species concentrations conc_nc with shape (n_nc, num_target_waters).
 - Allocate the kinetic reaction rates rk_out with shape (n_aq_kin + n_min_kin, num_target_waters).
- **Process:**
 - **Read $\tilde{\mathbf{U}}^{k+1}$ from file_in. Note:** The rows must represent components and the columns targets.
 - **Solve** reactive mixing iteration for each target water j :
 - **Loop over number of target waters**
 - Compute chemical reactions contribution to aqueous component concentrations using **Euler explicit** and **lumping**:
$$\mathbf{u}_{K,j}^{k+1} := \Delta t^k \mathbf{W} \mathbf{S}_{K,v}^T \mathbf{r}_{K,j}^k$$

- Update the total aqueous component concentrations in target water: $\mathbf{u}_j^{k+1} = \tilde{\mathbf{u}}_j^{k+1} + \mathbf{u}_{K,j}^{k+1}$.
- Speciate to compute variable activity species concentrations $\mathbf{c}_{v,j}^{k+1}$ from components, using Newton-Raphson.
- Check Newton convergence: if speciation does not converge for a target water, the target index and a "No convergence in speciation after reactive mixing iteration. Try reducing the time step" message are printed and execution is aborted with an error stop.
- Capture the kinetic reaction rates $\mathbf{rk_out}$ for the current target water from the aqueous and mineral kinetic rate arrays of the chemistry object.

- **Post-process:**

- **Write** $\mathbf{U}^{k+1}$ in `file_out` (same dimensions as $\tilde{\mathbf{U}}^{k+1}$)
- **Write** $\mathbf{C}_v^{k+1}$ in `file_out` (rows: species, columns: targets)
- **Write** kinetic reaction rates $\mathbf{rk_out}$ in `file_out` (rows: reactions [aqueous (linear+redox) | minerals], columns: targets)
- **Deallocate** arrays

interfaz_esp_arch

- **Description:** Subroutine that solves a reactive mixing iteration with **aqueous species**, and writes aqueous species concentrations and kinetic reaction rates, using **files** as arguments. The integration method is **Euler explicit** and the reaction mixing ratios are **lumped**. It assumes that there are no equilibrium reactions.
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
path	String	Path to input and output files.
num_aq_comps	Integer	Nº aqueous species (= nº aqueous components)
file_in	String	File containing concentrations of aqueous species after solving a conservative transport iteration ($\tilde{\mathbf{C}}^{k+1}$)
Delta_t	Real	Time step (Δt^k)
file_out	String	File where aqueous species concentrations ($\mathbf{C}^{k+1}$) and kinetic reaction rates after solving a reactive mixing iteration will be written

- **Pre-process:**
 - Allocate aqueous species concentrations after conservative transport ($\tilde{\mathbf{C}}^{k+1}$)
 - Allocate chemical reactions contribution to aqueous species concentrations ($\mathbf{C}_K^{k+1}$)
 - Determine, from the first target water, the number of kinetic reactions (aqueous kinetic n_aq_kin, mineral kinetic n_min_kin).
 - Allocate the kinetic reaction rates **rk_out** with shape (n_aq_kin + n_min_kin, num_target_waters).
- **Process:**
 - **Read** $\tilde{\mathbf{C}}^{k+1}$ from file_in. **Note:** The rows must represent species and the columns targets.
 - **Solve** reactive mixing iteration for each target water j :
 - **Loop over number of target waters**
 - Compute chemical reactions contribution to aqueous species concentrations using **Euler explicit** and **lumping**:
$$\mathbf{c}_{K,j}^{k+1} := \Delta t^k \mathbf{S}_{K,aq}^T \mathbf{r}_{K,j}^k$$
 - Update the total aqueous species concentrations in target water:
$$\mathbf{c}_j^{k+1} = \tilde{\mathbf{c}}_j^{k+1} + \mathbf{c}_{K,j}^{k+1}$$
 - Capture the kinetic reaction rates **rk_out** for the current target water from the aqueous and mineral kinetic rate arrays of the chemistry object.
- **Post-process:**

- **Write** C^{k+1} in `file_out` (rows: species, columns: targets)
- **Write** kinetic reaction rates `rk_out` in `file_out` (rows: reactions [aqueous (linear+redox) | minerals], columns: targets)
- **Deallocate** arrays